CLINICAL PRESENTATION

# Frailty Fractures and Functional Recovery

Comprehensive overview of common fracture patterns, prevention cascades, and geriatric rehabilitation

**By Dr Clement Vikash Vasanth Raj | 10/11/26**



MODULE 01

# The Epidemiology of Fragility

Understanding the global demographic challenge and clinical burden

# Global Incidence



# 9 Million

Osteoporotic fractures occurring worldwide every single year

# Biological vs. Chronological Age



## Defining Frailty

Frailty reflects cumulative physiological decline and vulnerability across multiple systems, independent of calendar age.

**Evaluated via the Clinical Frailty Scale (CFS).**

**20–30%**

One-year post-fracture mortality rate in vulnerable seniors

MODULE 02

# Most Common Elderly Fractures

Detailed breakdown of primary high-risk skeletal sites

# Major Fracture Sites



## Hip (Proximal Femur)

Highest mortality and permanent loss of independent living.



## Vertebrae (Spine)

Often silent fractures causing chronic pain and kyphosis.



## Distal Radius (Wrist)

Early sentinel indicator of underlying bone fragility.

# Secondary Common Sites



## Proximal Humerus

Impacts upper limb independence following low-energy falls.



## Pelvic Fractures

Increasing fragility fractures posing conservative management challenges.



## Rib & Ankle

Impairing respiratory mechanics and baseline ambulation.

# Pathophysiology of Bone Loss

## Remodeling Imbalance

Osteoclast activity outpaces osteoblast formation, leading to severe microarchitectural decay compounded by age-related muscle sarcopenia.

### Key Risk Drivers

- Hormonal shifts & Vitamin D deficiency
- Chronic systemic inflammation (inflammaging)
- Sedentary lifestyle and immobility

MODULE 03

# Prevention of Fractures

Primary fall reduction, polypharmacy review, and secondary FLS models

# Primary Fall Prevention



## Home Safety Audits

Removing loose rugs, installing grab bars, improving lighting.



## Sensory Screenings

Routine vision, hearing, and podiatry/footwear evaluations.



## Balance Programs

Targeted community exercise classes like Tai Chi.

# Managing Polypharmacy



## Targeting FRIDs

Fall-Risk-Increasing Drugs (sedatives, hypnotics, anticholinergics) must be systematically reviewed and tapered to prevent orthostasis and sedation.



## Medication Review

Safeguarding balance by eliminating unnecessary psychotropics and sedatives.

# Fracture Liaison Services (FLS)



## Closing the Prevention Gap

Coordinator-based care models ensure every fragility fracture patient receives timely DXA screening and bone protection before hospital discharge.

**2x Risk**

Subsequent fracture risk doubles within one year following an initial break

# Bone Health Pharmacotherapy



## Antiresorptives

Bisphosphonates and denosumab to halt osteoclast breakdown.



## Anabolic Agents

Teriparatide and romosozumab to build new bone matrix.



## Micronutrients

Consistent daily calcium and vitamin D optimization.

MODULE 04

# Acute Management & Surgery

Orthogeriatric co-management and surgical timing imperatives

# The Golden Window for Surgery



## Why Urgency Matters

Performing surgical intervention within 24 to 48 hours dramatically minimizes complications like pressure ulcers, VTE, and pneumonia.

**< 48h**

Target timeframe for surgical fixation in frail older adults

# Orthogeriatric Co-Management



## Shared Care Wards

Joint medical-surgical rounds reducing inpatient stays.



## Delirium Protocol

Multicomponent non-pharmacological cognitive care.



## Pain Management

Multimodal analgesia avoiding heavy opioid sedation.

MODULE 05

# Functional Recovery & Rehab

Restoring independence, mobility, and community reintegration

# Rehabilitation Pillars



## Early Mobilization

Getting patients out of bed on postoperative Day 1.



## Resistance Training

Progressive loading to counter muscle atrophy.



## Gait Retraining

Safe assistive device progression and balance practice.

# Nutritional & Metabolic Support



## Fueling Tissue Healing

Surgical trauma triggers severe catabolism. High protein intake, micronutrient supplementation, and hydration are vital for muscle protein synthesis.



### Key Targets

- High daily protein targets
- Vitamin D and calcium repletion
- Malnutrition screening protocols

# Psychological & Caregiver Support



## Overcoming Fear of Falling

Post-fall anxiety restricts activity and delays functional recovery. Providing psychological encouragement and supporting family caregivers ensures enduring success.



## Caregiver Strain

Mitigating burnout through structured respite and discharge planning resources.

MODULE 06

# Summary & Future Outlook

Key takeaways and closing clinical consensus

# Core Clinical Takeaways



## Holistic Teamwork

Orthogeriatric integration saves lives and function.



## Aggressive Prevention

FLS models successfully stop secondary fracture cascades.



## Early Mobility

The fundamental cornerstone of successful recovery.

**“Restoring mobility and independence after a fragility fracture requires synchronized acute care, proactive secondary prevention, and lifelong rehabilitation.”**

CLINICAL CONSENSUS STATEMENT



# Questions and Discussion

Thank you for your attention